भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS351: IT Workshop II (Cloud Computing), End Semester Examination, Monsoon 2025

Note: Maximum time: 3 Hours. Total Marks: 60

1. (a) [4 points] You need to build a web application for a multi-user event management system. How would you develop this application using Django?
   (b) [2 points] Service integration is an essential component in the design of Platform-as-a-Service (PaaS) solutions. Explain.

2. (a) [4 points] **Fill in the blanks.** Given a collection of documents, we implement a MapReduce solution to count the frequency of each word across all documents. The functions will be called in the following sequence: map_phase() → shuffle_sort() → reduce_phase()

```
def map_phase(doc):
    words = doc.split()
    return [(word, 1) for word in words]

def reduce_phase(grouped_data):
    word, counts = grouped_data
    return (word, sum(counts))

def shuffle_sort(mapped_data):
    grouped_data = __(a)_____________________
    for __(b)_____ in mapped_data:
        __(c)_____.__(d)_____
    return __(d)_____
```

   (b) [2 points] Is random read and write access allowed in HBase? Explain.

3. (a) [3 points] Explain the differences between the ResourceManager and the NodeManager in a YARN-based Hadoop cluster.
   (b) [3 points] You have n cloud tasks, and each task i has a reward that reflects how much it benefits from being allocated to a resource. Your objective is to maximize the total reward by choosing a set of tasks to allocate, subject to the constraint that no two consecutive tasks can be allocated to the same resource. Write an algorithm to solve this problem.

4. (a) [4 points] Write HiveQL statement to find the average salary of employees in each department, but only for employees whose salary is greater than the average salary for their own department.
   (b) [2 points] Can Pig Execution Engine execute the Pig Latin scripts by compiling them into MapReduce jobs?

5. (a) [3 points] How does Spark Streaming ensure fault tolerance? Explain.
   (b) [3 points] How is the data integrity managed in cloud computing? Give an example.

6. (a) [3 points] In Security Assertion Markup Language (SAML) based authentication, who performs the user authentication? Explain with an example.
   (b) [3 points] Can Digital Signatures guarantee confidentiality? Justify.

7. (a) [3 points] How Message Authentication Code prevents Data Tampering? Explain with an example.
   (b) [3 points] How can we implement K-Nearest Neighbors Algorithm using Map Reduce? Write map and reduce functions.

8. (a) [4 points] Explain how Multimedia Cloud Infrastructure can perform Video Encoding and Media Analytics.
   (b) [2 points] What mechanisms are used to ensure low latency during live streaming?

9. (a) [3 points] Write a Dockerfile for a python application. Assume that the python application will run in port 8080.
   (b) [3 points] We want to run a container for a Python app, and another container for a Redis database. Both containers need to communicate with each other. Write Docker commands to create a bridge network to enable this communication.

10. (a) [3 points] Communication between services over a network can introduce latency compared to monolithic architectures. Justify.
    (b) [3 points] Why do Microservices need a Message Broker for communication?